

Understanding Slope and Intercept ● Beginner

Turning the abstract b_0 and b_1 into something you can say out loud in a meeting.

WORKED EXAMPLE

Salary = 20,000 + 5,000 × Years of Experience

Intercept = 20,000. Slope = 5,000.

HOW TO SAY IT IN A SENTENCE

"For every additional year of experience, predicted salary increases by approximately ₹5,000, according to this model." Notice the careful phrasing — **according to this model**, not "experience causes higher salary." Association is not causation: confounding factors (seniority, company size, negotiation skill) could be driving both.

Positive, negative, and zero slope — interpreted

Slope	Interpretation
Positive (e.g. +5,000)	Outcome increases as the feature increases
Negative (e.g. -1,200)	Outcome decreases as the feature increases
Near zero	This feature has little linear predictive effect on the outcome, holding the model as-is

WHY THE INTERCEPT CAN BE PRACTICALLY MEANINGLESS

The intercept is the predicted value when every feature equals zero. For a house-price model, that would mean "a house with 0 square feet" — mathematically necessary for the line to exist, but not a real scenario. The intercept is often a mathematical anchor, not a real-world data point.